

Multiple Access Techniques

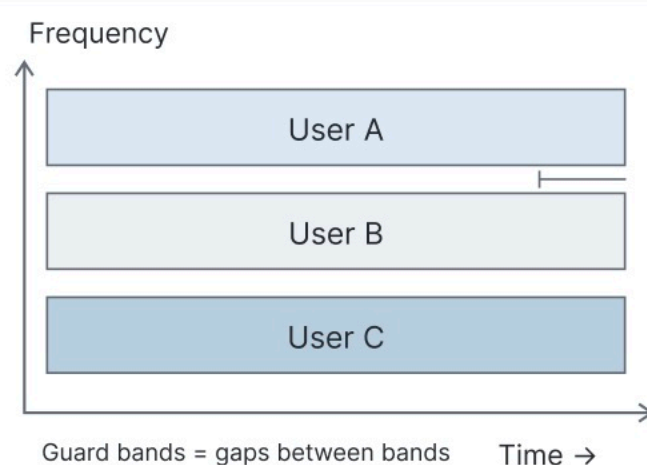
How multiple users share a communication channel

Same three users in every diagram

User A
 User B
 User C

FDMA

Frequency Division Multiple Access



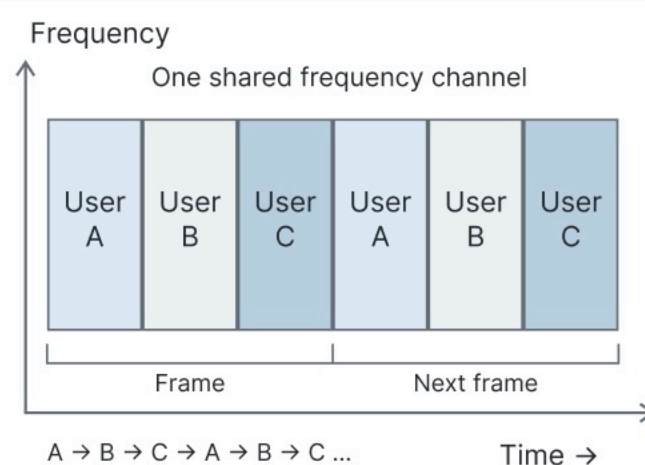
Users transmit at the same time in separate frequency bands.

KEY DIFFERENCE

One fixed band per user. Guard bands help reduce interference.

TDMA

Time Division Multiple Access



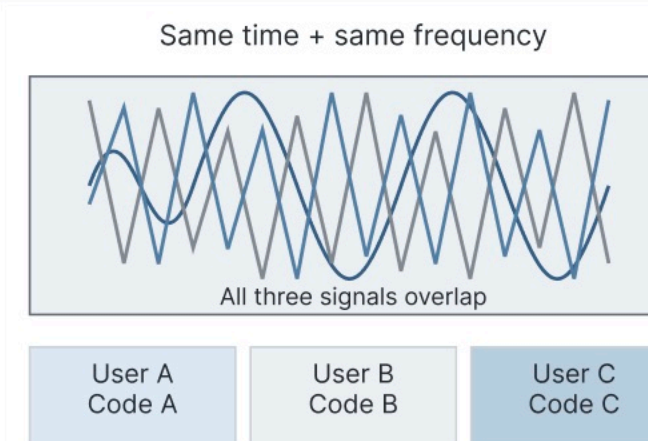
Users take turns in time slots on a shared frequency channel.

KEY DIFFERENCE

One user per slot. Users must synchronize their timing.

CDMA

Code Division Multiple Access



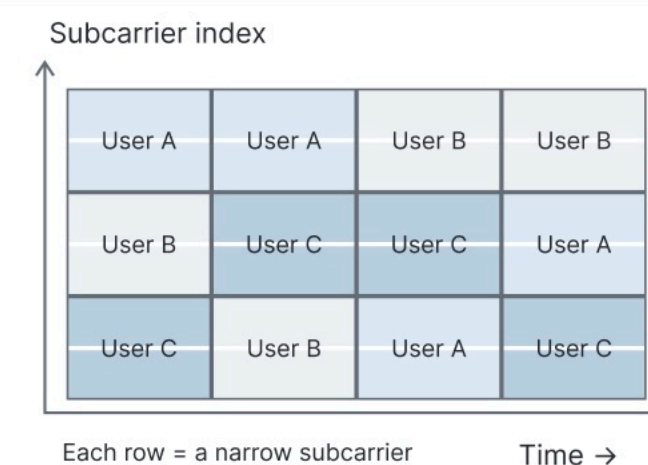
Users share a channel; the receiver separates signals using codes.

KEY DIFFERENCE

Same time and frequency, but a different spreading code per user.

OFDMA

Orthogonal Frequency Division Multiple Access



Users get groups of subcarriers during scheduled time intervals.

KEY DIFFERENCE

Unlike FDMA: many narrow orthogonal subcarriers, flexibly reassigned over time.

At a glance	FDMA	TDMA	CDMA	OFDMA
Separated by	Frequency bands	Time slots	Spreading codes	Subcarriers + scheduled time
Shared resource	Time	Frequency channel	Time and frequency	Time–frequency resource grid
Familiar example	Analog 1G (e.g. AMPS)	GSM combines TDMA with FDMA	IS-95 / cdma2000	LTE downlink / 5G NR